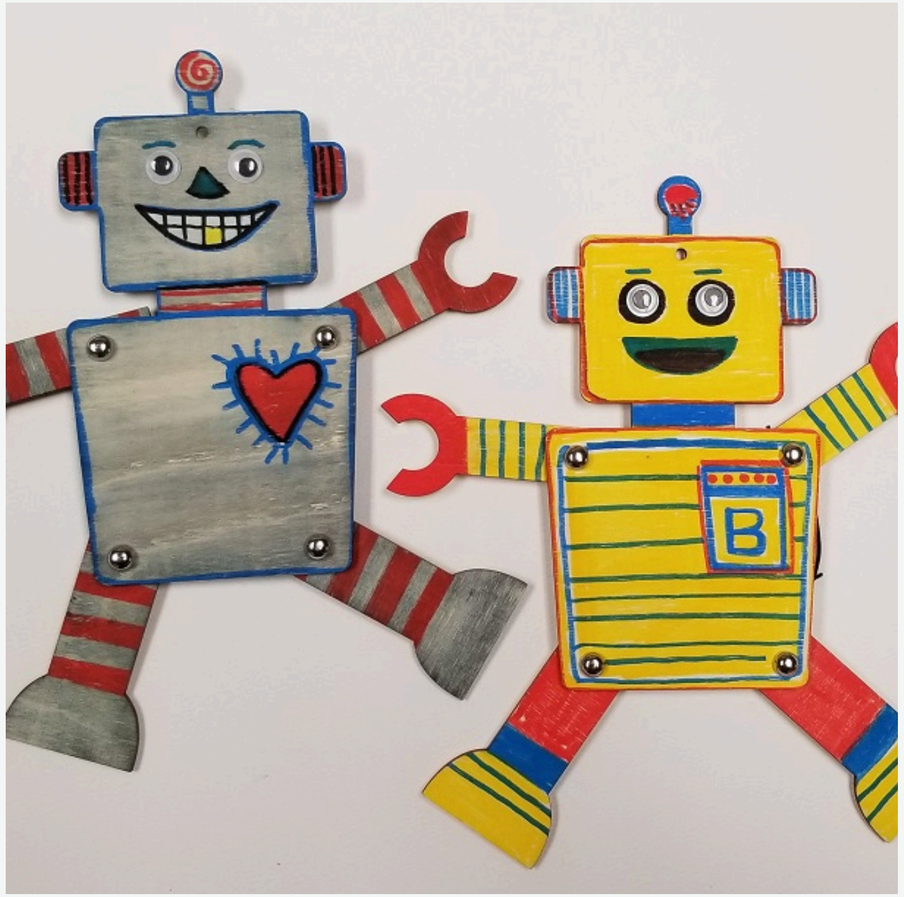
Lesson 10

While Loop

A While Loop is used when you want the computer to repeat a block of code over and over again as long as a condition remains True.

Think of it like a game: "While the player has lives > 0, keep the level running!"



How to Write a While Loop

```
x = 1
while x <= 5:
    print(x)
    x = x + 1
```

Start with the keyword `while`.

Add a condition and a colon `:`.

Indent the instructions below.

Crucial: Update the value (like x + 1) so the loop eventually stops!

While Loop

Lets say I want to print the number 5, 100 times. That would take way too long so we could instead use while loops

Logic

```
Condition = Question

Is timer > 0 ?

YES → keep looping

NO → stop
```

Quiz: Loops

Question 1

What happens if the condition is `False` the very first time Python looks at it?

Answer: The loop never starts! Python skips it entirely.

Question 2

What is an **"Infinite Loop"** and how do you accidentally create one?

Answer: A loop that never stops. This happens if the condition is always True.



Activity: Countdown Timer

Create a program that counts down from 10 to 1.

1. Create a variable called “timer” set to 10.
2. While the timer is greater than 0, print the number.
3. Subtract 1 from the timer each time.
4. Print "Blast Off!" at the very end.



Answer: Countdown Timer

```
# 1. Create a variable called timer set to 10
timer = 10

# 2. While the timer is greater than 0, keep looping
while timer > 0:
    print(timer)
     # 3. Subtract 1 from the timer each time to avoid an infinite loop
    timer = timer - 1  # or timer -= 1

# 4. Print "Blast Off!" at the very end outside the loop
print("Blast Off!")
```

Empty String

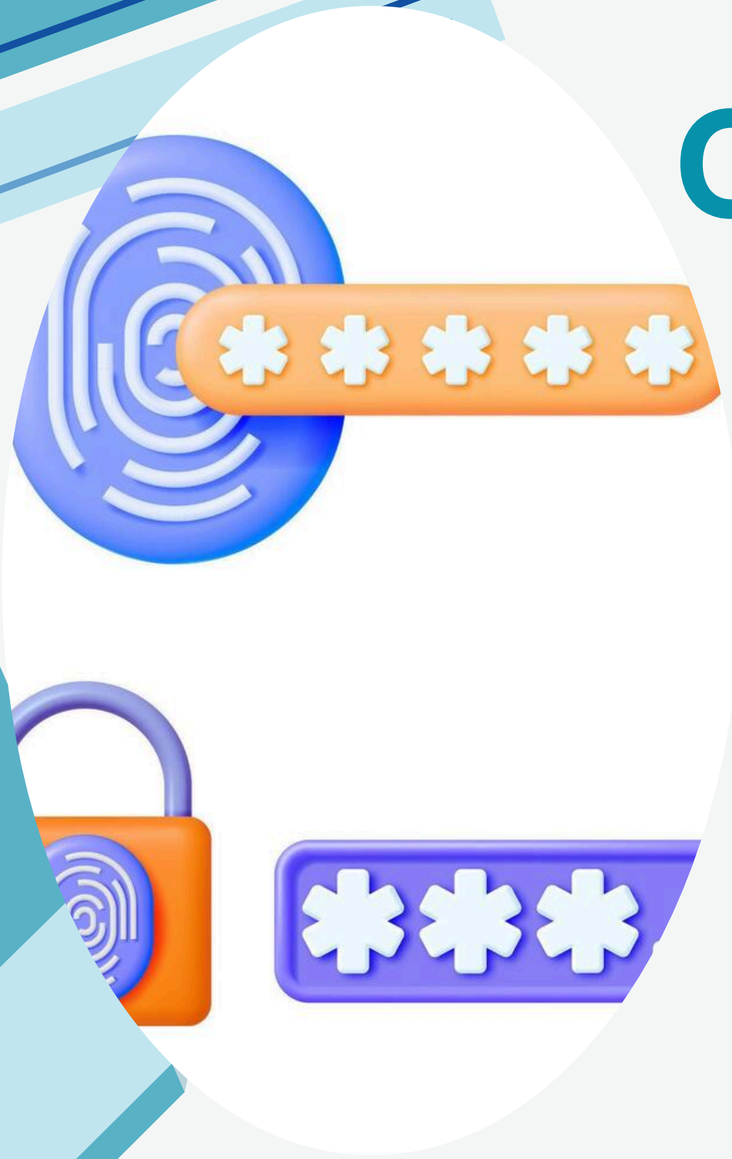
A Digital "Clean Slate"

An empty string `""` is a string that exists but has **zero characters** inside it.

- **Placeholder:** Use it to "reserve a spot" for text you will collect later.
- **Initialization:** Perfect for starting a variable before a `while` loop begins.
- **Comparison:** Unlike a number 0, it is specifically for **text** logic.



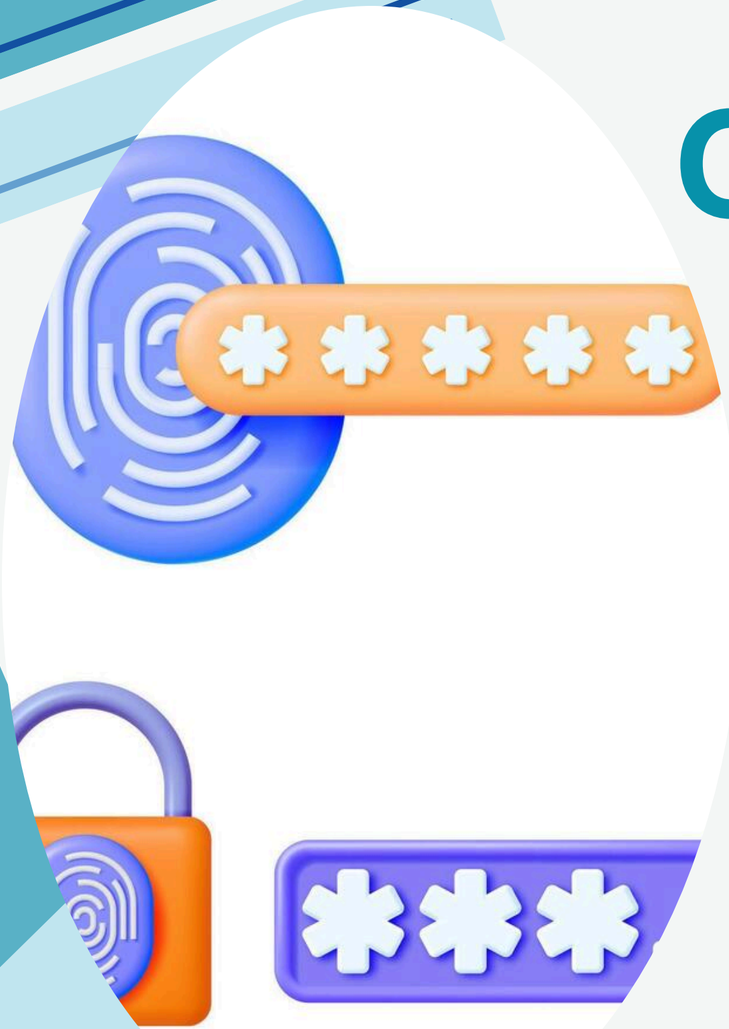
Activity: Countdown Timer



Write a loop that forces the user to keep guessing a password until they get it right!

- 1. Set a variable “guess” to an empty string "".
- 2. Use a loop: while guess != "CyberIQ":
- 3. Inside, ask for the password using input().
- 4. Only print "Access Granted" **outside the loop!**

Answer: Countdown Timer



```
# 1. Set a variable guess to an empty string ""
guess = ""

# 2. Use a loop that keeps going while the guess is wrong
while guess != "CyberIQ":
    # 3. Inside, ask for the password using input()
    guess = input("Enter the secret passcode: ")

# 4. Only print "Access Granted" outside the loop!
print("Access Granted")
```

While Loop Recap

1. Logic First

The loop checks the condition **before** running. If it's False at the start, the code never runs!

2. The Block

All code that belongs inside the loop **must be indented**. Python uses this space to know what to repeat.

3. The Update

You **must** change a variable inside the loop so the condition eventually becomes False and the loop stops.

Practice programs

Print 200 , 5s using while loops

so the number 5 should be printed out exactly 200 times, 1 on each line.

Keep asking user for numbers until the sum of all entered numbers hits 0.

Make a guessing game: select a number to be the secret number. Keep asking user for guesses until they get it right.